

08 — Mu-Calculus

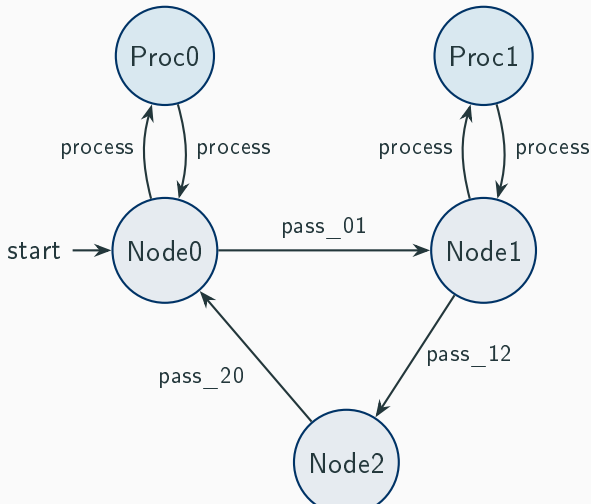
Fixpoints, Modalities, and Common Patterns

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Mununu Tutorial

The Token Ring

A token circulates among three nodes; each can **process** locally before passing it on.



Operator Reference

Syntax	Name	Meaning
$\text{nu } X. \varphi$	Greatest fixpoint	Safety / invariance
$\text{mu } X. \varphi$	Least fixpoint	Reachability / liveness
$[] \varphi$	Box	<i>All</i> successors satisfy φ
$\langle \rangle \varphi$	Diamond	<i>Some</i> successor satisfies φ
$[\text{ labels}=\{a\}] \varphi$	Labeled box	All <i>a</i> -successors satisfy φ
$\langle \text{ labels}=\{a\} \rangle \varphi$	Labeled diamond	Some <i>a</i> -successor satisfies φ
$[(\text{ctrl}=\text{controllable})] \varphi$	Ctrl box	All uncontrollable successors <i>and</i> some controllable successor satisfy φ

Safety and Reachability

```
1 // == Greatest fixpoint (nu) --Safety/Invariance ==
2
3 // nu X. ([] X) --"all reachable states are valid"
4 // The canonical safety invariant.
5 formula safety_invariant {
6     over TokenRing;
7     body = nu X. ([] X);
8 }
9
10 // == Least fixpoint (mu) --Reachability ==
11
12 // mu X. (Node2 || <> X) --"Node2 is reachable from here"
13 // Computes the set of states from which Node2 can be reached.
14 formula node2_reachable {
15     over TokenRing;
16     body = mu X. (Node2 || <> X);
17 }
```

Labeled Modalities

```
1 // == Labeled box modality ==
2
3 // [ labels = { pass_01 } ] Node1
4 // "Every pass_01 transition leads to Node1"
5 // True for all states (pass_01 only exists from Node0 -> Node1).
6 formula pass01_leads_to_node1 {
7     over TokenRing;
8     body = [ labels = { pass_01 } ] Node1;
9 }
10
11 // == Labeled diamond modality ==
12
13 // < labels = { pass_12 } > Node2
14 // "There exists a pass_12 transition leading to Node2"
15 // True only for Node1 (the only state with a pass_12 transition).
16 formula can_pass_to_node2 {
17     over TokenRing;
18     body = < labels = { pass_12 } > Node2;
```

Nested Fixpoints — GF Pattern

```
1      // === Nested fixpoints --GF pattern ===
2
3      // nu X. (mu Y. (Node0 || <> Y)) && ([] X)
4      // "Infinitely often Node0" (GF Node0)
5      // From every reachable state, Node0 can always be reached again.
6      formula gf_node0 {
7          over TokenRing;
8          body = nu X. ((mu Y. (Node0 || <> Y)) && ([] X));
9      }
```

$\nu X. (\mu Y. (\text{Node0} \vee \Diamond Y)) \wedge []X$

- **Outer** ν (safety): the property holds at every reachable state.
- **Inner** μ (reachability): Node0 can be reached from the current state.
- **Combined**: “from every reachable state, Node0 can always be reached again” — the **GF (infinitely often)** pattern.

Common Patterns Summary

Pattern	Formula
Safety invariant	$\nu X. []X$
Reachability	$\mu X. (p \vee \Diamond X)$
Labeled transition check	$[labels = \{a\}] q$ or $\langle labels = \{a\} \rangle q$
GF (infinitely often)	$\nu X. (\mu Y. (p \vee \Diamond Y)) \wedge []X$
No deadlock	$\nu X. \Diamond \text{true} \wedge []X$

Key insight:

- ν (nu) starts from “everything satisfies” and removes violations — **safety**.
- μ (mu) starts from “nothing satisfies” and adds witnesses — **reachability**.
- Nesting them yields **liveness** properties (e.g., infinitely often).